

# Akashdeep Majumder

## Frontend Engineer

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[Portfolio](#) | [linkedin.com/in/akashdeep-majumder](https://linkedin.com/in/akashdeep-majumder) | [github.com/akashdeep-majumder](https://github.com/akashdeep-majumder)

Frontend engineer with 3 years of sole ownership over a production robotics platform, making architectural decisions across real-time data pipelines, 3D rendering, and state management. Diagnosed and resolved a 5 FPS rendering problem as two independent engineering issues, not one. Also designed and shipped a full-stack social platform using Next.js and TypeScript.

## SKILLS

**Frontend:** React.js (v18/v19), Next.js (App Router, SSR, Server Components), JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS

**3D & Visualization:** Three.js, React Three Fiber, Recharts

**State Management:** Redux Toolkit, Redux Thunk, React Context API

**UI & Styling:** Material UI, Tailwind CSS, Styled Components, CSS animations

**Performance:** Core Web Vitals, Lighthouse, Chrome DevTools, critical rendering path, memory profiling

**APIs & Auth:** REST APIs, WebSocket, JWT, Axios

**Testing & DevOps:** Jest, Vitest, React Testing Library, Git, CI/CD, Vite, Docker, Vercel, Render

**Backend:** Node.js, Express.js, FastAPI, Python

**Database:** MongoDB, MySQL

## EXPERIENCE

**Software Engineer** | Thedush Robotics

Mar 2023 - Apr 2026

Sole frontend engineer on a production robotics telemetry and visualization platform. Owned every architectural decision, from rendering pipeline choices to state management patterns, across 3 years.

### ONIX - Robot Telemetry & Visualization

- Identified Canvas 2D as the wrong tool for large warehouse maps with dense coordinate data and migrated the LiDAR visualizer to React Three Fiber. GPU rendering handled spatial complexity Canvas physically couldn't sustain.
- After the migration, frame rate was still stuck at 5 FPS. Profiled the pipeline and traced it to bandwidth, not rendering. Evaluated WebSocket against WebRTC, chose WebSocket with Float16 and Deflate compression. Bandwidth dropped from 1.5 MB to 100 KB per frame, throughput rose to 50+ FPS.
- Designed a centralized map management system with a normalized schema after recognizing that each feature maintaining its own map structure was the source of recurring, hard-to-trace bugs. Eliminated that category of failure.
- Engineered an Immer patch-based undo-redo system, choosing diff-based history over full snapshots after profiling showed memory bloating on large maps with many operations. Cut memory overhead by roughly 90 percent. Mentored 5-6 interns through code reviews over 3 years.

### WMS + Robot Integration

- Architected and delivered a full Warehouse Management System for the Indian Army in 14 days, integrating robots and Pick-to-Light guidance. Automated retrieval cut picking time roughly in half. Packaged as an air-gapped offline desktop app for factory-floor deployment.

Process: daily standups, code reviews, weekly sprint reviews.

## PROJECTS

### Adda - Social Media Platform

2026

- Designed and built a full-stack social platform with Next.js 16 App Router and Server Components, choosing SSR for authenticated feeds and SSG for public pages based on data freshness requirements. TypeScript interfaces enforce API contracts throughout.
- JWT stored in HTTP-only cookies with sameSite configuration for cross-domain deployment. Zod validates every API route. Core Web Vitals optimized to green via Lighthouse.

Stack: Next.js 16, React.js, TypeScript, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, JWT, Zod

See more projects (Robot Config Builder, Network Strength Visualizer, Cricket Scoring App): [akashdeep-majumder.github.io](https://akashdeep-majumder.github.io)

## EDUCATION

**Bachelor of Science (Graduate)** | Tripura University

2019 - 2022